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Q3 Indeed, the accuracy of classification finding is greatly impacted by the quality of segmentation. A crucial first step in ~~segmentation~~ segment-based image analysis is precise segmentation, and when done correctly, it guarantees that the best Segmentation Outcome is applied to image Classification.

An example of from a study on (medical ~~image~~ imaging) (tumor Classification in Cancer research)

- ① Inadequate or poor Segmentation: The retrieved features could be deceptive if Segmentation leaves out important adjacent tissues or does not capture the full tumor.
- ② High quality Segmentation: The classifier may concentrate on the pertinent properties (such as form, tissue and intensity) to successfully differentiate between benign and malignant tumors if the Segmentation correctly captures the tumor region without adding superfluous background.

Q2

function modify_image(image):

Let's assume 'image' is a 2D array of size $m \times n$ representing a grayscale image

Where pixel values ranges from 0-255

Let's find the min and max pixel values in the image

min_value = minimum value in image

max_value = maximum value in image

Let's create a new output image of the same size as the input

Output = new 2D array of size $m \times n$

Let's iterate through each pixel in the image

for i from 0 to $m-1$:

for j from 0 to $n-1$:

if image[i][j] == max_value:

The criterion i: Brightest pixel(s) become darkest
Output[i][j] = 0

else if image[i][j] == min_value:

The criterion ii: Darkest pixel(s) become Brightest
Output[i][j] = 255

else if $175 \leq \text{image}[i][j] \leq 200$:

The criterion iii: Pixels between 175 and 200 become 120

$$\text{Output}[i][j] = 120$$

else:

keep other pixels unchanged

$$\text{Output}[i][j] = \text{image}[i][j]$$

return output

Q1 The 3 bit quantization

(i) An 8-bit image has pixel values in the range (0 to 255)

(ii) 3-bit image range 0 to 7 (i.e. $2^3 = 8$)

(iii) 8-bit values (0-255) and 3-bit values (0-7)

The Quantitative mapping

Here, x is the original pixel value (0-255)
and since $(256/8 = 32)$

For each pixel in the given matrix

$$Q(x) = \left\lfloor \frac{x}{32} \right\rfloor$$

original	calculation	quantized (3-bit)
188	$188/32 = 5.875$	5
91	$91/32 = 2.84375$	2
162	$162/32 = 5.0625$	5
11	$11/32 = 0.34375$	0
130	$130/32 = 4.0625$	4
57	$57/32 = 1.78125$	1
112	$112/32 = 3.5$	3
111	$111/32 = 3.46875$	3

103	$103/32 = 3.21875$	3
187	$187/32 = 5.84375$	5
25	$25/32 = 0.78125$	0
40	$40/32 = 1.25$	1
220	$220/32 = 6.875$	6
20	$20/32 = 0.625$	0
220	$220/32 = 6.875$	6
203	$203/32 = 6.34375$	6

(4)

Quantized 3-bit values

$$\begin{bmatrix} 5 & 2 & 5 & 0 \\ 4 & 1 & 3 & 3 \\ 3 & 5 & 0 & 1 \\ 6 & 0 & 6 & 6 \end{bmatrix}$$

There, this is the 3-bit quantization output image.

Q4

(i)

In row-major order, elements are stored row by row, i.e

Given 3×3 matrix

$$\begin{bmatrix} 88 & 91 & 62 \\ 130 & 78 & 112 \\ 43 & 187 & 5 \end{bmatrix}$$

The row-major order representation is

$$\{88, 91, 62, 130, 78, 112, 43, 187, 5\}$$

(5)

(5)

Nodes: Each pixel (x, y) in image is a node in the graph. let's label the nodes as $P(x, y)$

$$P(0, 0) = 88, P(0, 1) = 91, P(0, 2) = 62$$

$$P(1, 0) = 130, P(1, 1) = 78, P(1, 2) = 112$$

$$P(2, 0) = 43, P(2, 1) = 187, P(2, 2) = 5$$

i.e $\Rightarrow$ nodes: 88, 91, 62, 130, 78, 112, 43, 187, 5

Edges: If a neighbor falls outside the image boundaries, there's no image. Because we connect each pixel to its 4-neighbors i.e (horizontally & vertically)

~~$P(0, 0)$~~

$$88 \leftrightarrow 91, 91 \leftrightarrow 62 \text{ (horizontal)}$$

$$88 \leftrightarrow 130, 91 \leftrightarrow 78, 62 \leftrightarrow 112 \text{ (vertical)}$$

$$130 \leftrightarrow 78, 78 \leftrightarrow 112 \text{ (horizontal)}$$

$$130 \leftrightarrow 43, 78 \leftrightarrow 187, 112 \leftrightarrow 5 \text{ (vertical)}$$

$$43 \leftrightarrow 187, 187 \leftrightarrow 5 \text{ (horizontal)}$$

(5)

(6)

(i)

The histogram of the image: The histogram represents the frequency of each intensity level in the image. Since $k_{\text{Max}} = 7$, the intensity levels range from 0 to 7.

0 : 1

1 : 6

2 : 3

3 : 2

4 : 3

5 : 2

6 : 1

7 : 2

Histogram representation

Intensity	frequency
0	1
1	6
2	3
3	2
4	3
5	2
6	1
7	2

(ii)

No, the image does not have high Contrast.

Justification: Contrast refers to the difference in intensity levels in an image.

(i) The intensity values range from 0 to 7, which is a relatively narrow range.

(ii) Many pixels have similar intensity values (i.e. a high frequency of ~~low~~ "1")

(6ii) There aren't large changes in intensity

(7)

(iii)

Histogram of the original image:

intensity	frequency
0	1
1	6
2	3
3	2
4	3
5	2
6	1
7	2

Total pixel: $4 \times 5 = 20$

Normalized histogram: let's divide each frequency by the total number of pixels 20

CDF: let's calculate the cumulative sum of the normalized histogram

intensity	frequency	normalized histogram ($f/20$)	CDF
0	1	0.05	0.05
1	6	0.30	0.35
2	3	0.15	0.50
3	2	0.10	0.60
4	3	0.15	0.75
5	2	0.10	0.85
6	1	0.05	0.90
7	2	0.10	1.00

Let's calculate the Equalized intensity values

Here, we multiply each CDF value by $K_{Max}(7)$ and round to two decimal places. ~~the~~ which will map out the old intensity values to new values.

old intensity	CDF	CDF $\times 7$	new intensity
0	0.05	0.35	0
1	0.30	2.10	2
2	0.15	1.05	3
3	0.10	0.70	4
4	0.15	1.05	5
5	0.10	0.70	6
6	0.05	0.35	6
7	0.10	0.70	7

Let's map Old Intensities to New Intensities in the image

Here, we will replace each pixel value in the original image with its corresponding new intensity value,

Equalized Image:

0	2	2	4	5
7	3	6	6	7
6	4	3	2	2
2	5	5	3	2

Updated Equalized Histogram

Here, we need to count how many times each new intensity value appears in the equalized image.

0: 1

2: 5

3: 2

4: 2

5: 2

6: 3

7: 2

Intensity	frequency
0	1
2	5
3	2
4	2
5	2
6	3
7	2